HAN LI

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Database Systems · Distributed Systems · Concurrency Control

EDUCATION

University at Buffalo, SUNY Sep 2024 - Present

Ph.D. in Computer Science and Engineering · Advisor: Prof. Haonan Lu

Tongji University Sep 2015 - Jul 2019

B.Eng. in Software Engineering

RESEARCH EXPERIENCE

Juicer: Proactive Ordering & Conflict Pacing Sep 2024 - Present

Research Software Engineer · Planned submission: OSDI 2027

- Exploring whether **message order and timing** can improve existing concurrency control: cross-shard order inversions cause aborts, but conflicting messages can also cause aborts without inversions.
- Developing timestamp ordering and dependency-guided pacing at hotspots; prototyping in CockroachDB and Spanner/NCC research implementations. Using per-key abort attribution to evaluate avoided retries against added waiting and end-to-end latency.

Lakehouse Concurrency Control Apr 2026 - Present

Research Software Engineer · Planned submission: OSDI 2027

- Investigating why small write conflicts trigger full job retries after expensive computation and object-store I/O. The hypothesis: repair should scale with **conflicting data**, not the entire transaction.
- Designing early intent-based ordering and incremental repair in Spark/Iceberg to reuse valid outputs and check changes against committed snapshots. Building concurrent SQL workloads, S3 tracing and serial replay for ongoing correctness and I/O evaluation.

INDUSTRY EXPERIENCE

Alibaba Group · Dataphin Aug 2019 - Jul 2022

Software Development Engineer II · Earlier internship: Jul 2018 - Jan 2019

- Built resource scheduling for **TB-scale memory resources**: implemented a Mesos framework and an asynchronous worker event loop/state machine; designed a subsequent Kubernetes-based scheduling solution.
- Led a **10-person cross-functional team** on a customer-tagging platform; translated tag configurations into dependent Hive/MaxCompute jobs and built SQL/UDF pipelines and materialized-view abstractions.
- Designed metadata discovery, RocketMQ-based delivery and search using Elasticsearch/PostgreSQL GIN indexes, with a unified interface for tables, columns, functions and workspaces.

Zhejiang Bujuan Tech Aug 2021 - Aug 2024

Co-founder

- Co-founded **Nie Le Ge Lian**, a WeChat mini-program community for creating and sharing original characters, artwork and stories.
- Built a data pipeline for client behavior, server logs and database records; designed collaborative-filtering training, online candidate retrieval and retrieval evaluation for content discovery.

TECHNICAL SKILLS

Languages: Java, Go, C++, Python, SQL
Data systems: Spark, Iceberg, Hive, Flink
Infrastructure: Kubernetes, AWS, Prometheus

Review draft